SI Appendix

Two geographies of urban deprivation: everyday services and emergency care across European cities

Tina Comes

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S1 Extended Materials and Methods

S1.1 Two meanings of unreachable

Unreachable conflates two situations that the pipeline keeps apart. A cell with no network path reaches zero facilities of any service in any mode; it is detected once by a service- and mode-independent probe, so the everyday and emergency masks are equal by construction, and it enters no statistic. A cell that is routable but has no facility of a given service within that service's cut-off is service-deprived rather than unreachable: it receives a large finite effective time (the routing cut-off) so that the everyday level saturates and the cell stays on the map. Merging the two would let one sparse service mask a whole city. Travel-time summaries (mean, median, 90th percentile) are computed over reachable cells only and reported beside the unreachable population share.

S1.2 Congestion and substitutability in the everyday regime

For each everyday service the two-step floating catchment area factor computes the supply-to-demand ratio of each facility, $R_j = S_j / \sum_i P_i K(t_{ij})$ with a Gaussian kernel K, and converts it into a time inflation $c_j = (R_{\text{ref}}/R_j)^{\gamma}$ relative to the demand-weighted median ratio of the city. The soft minimum over reachable facilities lies between the nearest adjusted time and that time less $\ln(n)/\kappa$. The everyday composite is the weighted mean of the per-service levels over the categories reachable from the cell; school and green-space sub-types share one extraction and weigh 0.5 each so that each of the five categories counts once. Because network walking time scales inversely with speed, the 3.5 km/h often assumed for elderly and mobility-impaired residents is equivalent to lowering t_0 from 15 to about 11 minutes at the baseline speed, which the 10-minute variant of the sensitivity grid covers.

S1.3 Anchor structure of the emergency cost

The 8-minute mark cannot play the half-maximum role the 15-minute inflection plays for the everyday level: convexity caps $g(8)/g(15)$ at 0.427 given $g(4)/g(15) = 0.10$, so the anchor structures of the two regimes are genuinely different. Free-flow civilian drive times anchor the cost scale and are not response-time predictions: dispatch and turnout are unmodelled, lights-and-siren travel is faster than civilian flow, and free-flow understates congested cores relative to the periphery.

S1.4 Vulnerability strata

Vulnerability is analysed on two strictly separated levels. The census-harmonised age strata (top quartile of the share under 15, of the share aged 65 and over) exist for every city and are pooled. National fine-grid strata (lowest rent and ownership quartiles, finer age cuts) exist only where a national grid does, use different definitions, and are reported per city and never pooled. Each stratum is compared against the cells on which its own variable is published, because national grids are published where the population is large enough and that is not a random subset of a metropolitan area. Within-stratum inequality uses the concentration index [1].

S1.5 Clustering and coverage grades

The city vector (means, Ginis, coupling, compounding, tail shares, within-city density slopes, log population) is robust-scaled, with the zero-inflated tail shares opened by $\log(1 + x)$, and clustered by k-means and Ward with k selected by silhouette over 2–6 [2]. Every partition carries a bootstrap adjusted Rand index over resampled cities [3], a Gaussian silhouette null (same-covariance multivariate-normal draws pushed through the same selection) and the ARI

against the region partition. When the winning split isolates an outlier group below a quarter of the sample, a peeled second pass re-scales and re-clusters the remaining cities with the same diagnostics. Desert is the main pass's outlier group, partial desert the peeled pass's smaller cluster, covered the rest. A feature-set check re-runs the whole procedure with the 8- and 15-minute benchmark shares added. The graded scaling model regresses log emergency cost on centred log population, grade dummies and their interaction with country-clustered errors.

S1.6 Deprivation against access

Every headline size regression is re-run on the deprivation-free travel-time indicators (median, mean and, for the emergency regime, the 90th-percentile minutes); the deprivation-based and time-based Gini families are correlated across cities; the desert capitals' ratios to the sample are computed under both measures. The everyday 90th percentile is not regressed because above the walking cut-off the everyday composite time behaves as a count of unmet service categories rather than a travel time.

S1.7 Robustness harness

A structured, non-probabilistic sweep recomputes the rank-based targets over a defensible envelope. Layer 1 varies curvature within the fixed forms. Layer 2 swaps the forms while pinning the same anchors: a concave Box-Cox everyday level through the same half-maximum and ceiling, and for the emergency regime an exponential escalation [4] and a bounded survival-based curve, both solved from the 8- and 15-minute anchors so that the three emergency forms coincide inside the benchmark window and diverge only beyond it. Layer 3 sweeps the accessibility model (soft-minimum κ, congestion exponent γ, catchment bandwidth, number of nearest facilities, everyday mode set, unreachable fill, off-network policy) from the saved travel-time matrices. A threshold axis sweeps the typology cut from the 40th to the 75th percentile.

S1.8 Reproducibility

The pipeline is configuration-driven with provenance sidecars on every derived travel-time matrix, and no deprivation parameter is hard-coded. Every number quoted in the article is checked against its source table by an automated audit of 190 checks run in continuous integration; the main-text figures are drawn from the pack tables and the configuration by scripts in the study repository.

S2 Deprivation vs access: exact values

Table S1: Size elasticities of deprivation-based and minutes-based outcomes (log-log, country-clustered SEs, 67 cities), plotted in main-text Fig. 2A. Source: `deprivation_vs_access.csv`.

Regime	Outcome	Elasticity	p	R^2
Everyday	mean deprivation level	−0.196	< 0.001	0.44
Everyday	median walk time	−0.237	< 0.001	0.16
Everyday	mean walk time	−0.178	< 0.001	0.31
Emergency	mean deprivation cost	−0.059	0.16	0.02
Emergency	median drive time	−0.043	0.09	0.04
Emergency	mean drive time	−0.043	0.05	0.04
Emergency	p90 drive time	−0.059	0.02	0.05

Table S2: The five emergency-desert capitals: median drive time to the nearest emergency facility and its ratio to the 67-city sample median (12 min) against mean emergency deprivation cost (multiples of g(15)) and its ratio to the sample value (1.14). Source: `desert_access_contrast.csv`.

City	Country	Median time (min)	Time ratio	Mean cost	Cost ratio
Vilnius	LT	14.0	1.17	5.39	4.72
Ljubljana	SI	23.0	1.92	4.96	4.34
Rīga	LV	13.5	1.13	3.37	2.95
Bucureşti	RO	8.5	0.71	3.21	2.81
Tallinn	EE	13.0	1.08	2.80	2.45

S3 Size regressions: diagnostics and the conditional emergency Gini

The everyday elasticity of -0.196 is stable under country fixed effects (-0.17), leave-one-out city deletion (envelope -0.204 to -0.185, Athina the most influential city), outlier-robust estimators (Huber -0.196, median regression -0.223) and exclusion of the five desert capitals (-0.203). Equivalence at the stricter ± 0.10 bound is not shown for the emergency elasticity, hence "bounded within ± 0.13".

The emergency-Gini size behaviour is conditional on the beyond-benchmark escalation form. Under the bounded survival-based curve the Gini is size-flat ($+0.001$, $p = 0.97$); under polynomial Box-Cox escalation, the published baseline, it is size-flat across the whole curvature sweep (-0.006 to $+0.014$, none significant), in agreement with the deprivation-free travel-time Gini (-0.015, $p = 0.42$); under exponential escalation it rises ($+0.098$, $p = 0.0003$), because exponential pricing turns the Gini into an index of extreme-tail depth (it saturates at a median of 0.96, decouples from the time distribution's inequality, $\rho = 0.04$, and tracks the beyond-45-minute population share, $\rho = 0.45$), and how many minutes past the cut-off the worst-served residents sit does grow with size even though the tail's population share does not (beyond-30-minute share elasticity -0.003, $p = 0.99$).

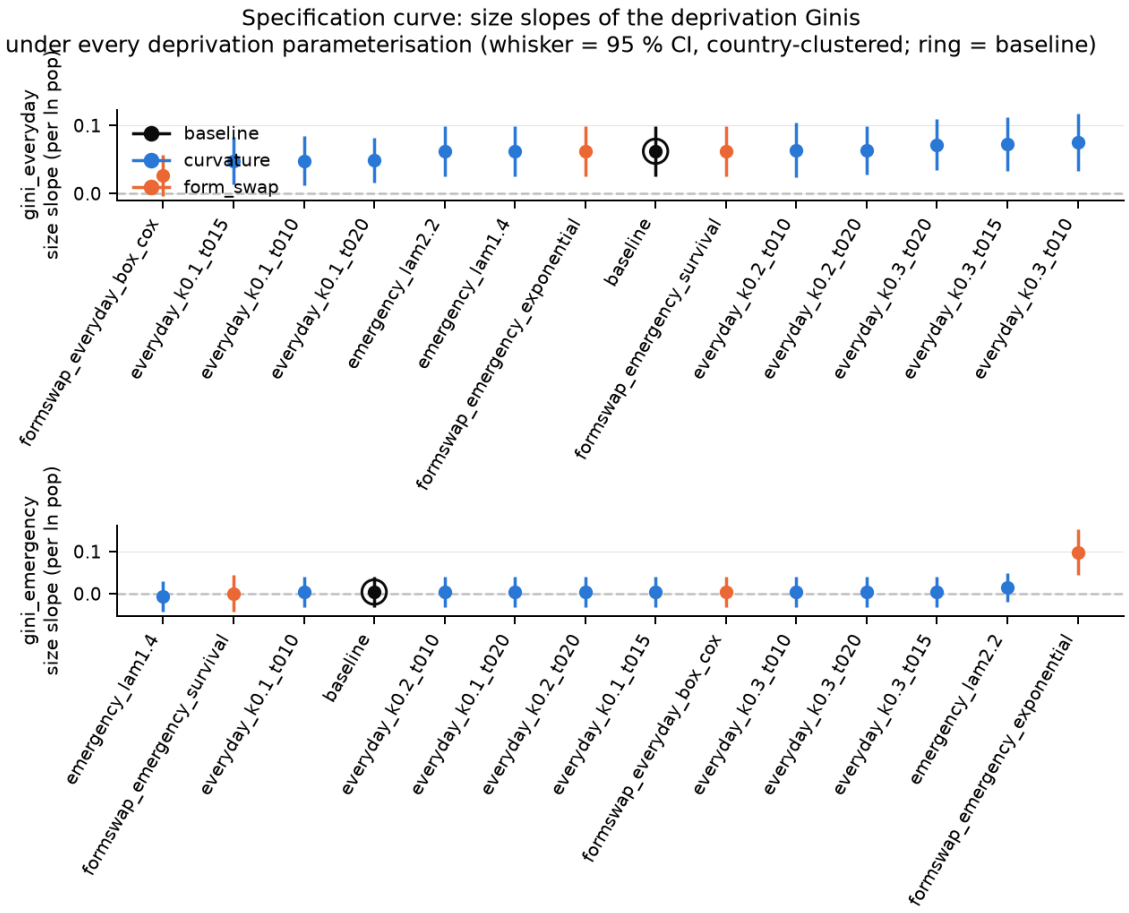


Figure S1: Specification curve of the inequality-scaling slopes. Top: everyday Gini; bottom: emergency Gini. Each point is the size elasticity of the within-city Gini re-estimated under the ringed baseline or one of the 14 alternative deprivation parameterisations (blue: curvature grid; orange: functional-form swaps); whiskers are country-clustered 95 % CIs. The everyday-Gini slope is positive under every specification (+0.026 to +0.075; significant in 13/14, the exception the concave Box-Cox form swap at $p = 0.083$). The emergency-Gini slope splits by beyond-cut-off escalation form.

S3.1 Additional robustness of the size regressions

Table S3 re-estimates the four headline size regressions with further checks. Residual spatial dependence is absent for the everyday level (Moran's $I = 0.06$, permutation $p = 0.41$) and present for the emergency level ($I = 0.23$, $p = 0.0015$) and for both Ginis ($I = 0.20$ and 0.39): cities close to one another share what size does not explain, which is the country and coverage structure of Section S4. The slope estimates are unaffected; inference is country-clustered throughout and regional contrasts permute countries, so the dependence enters the tests at the level at which it arises. Adding mapped GP density per 100k inhabitants as a control leaves the everyday elasticity at -0.185 (wild $p = 0.0002$) and the everyday-Gini elasticity at $+0.051$ ($p = 0.0006$); the control is significant in both everyday equations and in neither emergency equation. A centred quadratic in log population is not significant for the everyday level or either Gini and is significant for the emergency level (-0.169, wild $p = 0.001$; -0.104, $p = 0.010$ with coverage-grade dummies): the size-stratum means of the emergency deprivation cost are 1.15, 1.69, 1.56 and 0.70 multiples of g(15) from the smallest to the largest stratum, an inverted U in which the middle strata, where the desert and partial-desert cities sit, carry the highest cost and the four FUAs above five million the lowest. With identical regressors in both regime equations, joint estimation coincides with equation-by-equation OLS; the residual correlation it would add is $r = 0.51$ for the means and $r = 0.19$ for the Ginis ($p = 0.13$). The

minimum detectable emergency elasticity at 80 % power is 0.12.

Table S3: Additional robustness of the four log-log size regressions (67 cities, country-clustered). Moran's I of the residuals over the city-centre coordinates (5 nearest neighbours, row-standardised; expected value -0.015; permutation p, 9,999 draws); the elasticity with log mapped GP density per 100k inhabitants added as a control (wild cluster bootstrap p for the elasticity; clustered p for the control); the coefficient on the squared centred log population (wild p); the country-clustered Wald p of three size-stratum dummies replacing log population; and the minimum detectable elasticity at 80 % power (2.80× clustered SE). Source: `scaling_robustness_extra.csv` (`paper/analysis_scaling_robustness.py`).

Outcome	Elast.	Moran I	p_{perm}	Elast. GP	p_{wild}	GP coef. (p)	Quadr. (p_{wild})	Strata p
mean everyday level	-0.196	+0.061	0.406	-0.184	0.0002	-0.125 (0.00)	-0.023 (0.24)	0.000
mean emergency cost	-0.059	+0.232	0.001	-0.048	0.2838	-0.115 (0.18)	-0.169 (0.00)	0.000
everyday Gini	+0.062	+0.197	0.005	+0.051	0.0004	+0.121 (0.00)	+0.003 (0.76)	0.052
emergency Gini	+0.004	+0.388	0.000	+0.006	0.7998	-0.018 (0.47)	+0.004 (0.82)	0.759

Table S4: Cross-equation and cross-regime correlations. Residual correlation between the everyday and the emergency size regression (with identical regressors, joint SUR estimation coincides with OLS and this correlation is what joint estimation adds) and the cross-city correlation of the two within-city Ginis. Source: `scaling_crossequation.csv`.

Pair	r	p
residuals of the mean equations	+0.511	0.0000
residuals of the Gini equations	+0.186	0.1322
everyday Gini vs emergency Gini (Pearson)	+0.123	0.3202
everyday Gini vs emergency Gini (Spearman)	+0.117	0.3439

S4 Regional structure

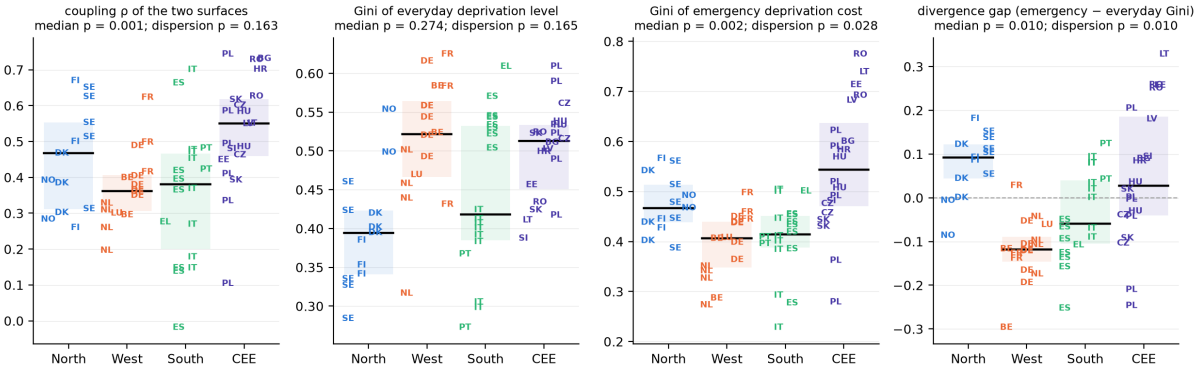


Figure S2: Regional structure of the divergence, each city drawn as its country code (bar: regional median; band: interquartile range) for the coupling ρ, the everyday Gini, the emergency Gini and the divergence gap. Panel titles give the country-permutation p-value of the regional contrast in medians and in dispersion. The everyday-Gini panel shows the Spanish and Greek cities above the Italian and Portuguese ones with no overlap, which is why that contrast is a country contrast and is not claimed regionally.

The West's inequality problem is everyday-side (median divergence gap -0.12); the North's is emergency-side ($+0.09$). The everyday Gini's apparent regional contrast has a city-level ANOVA $p = 0.0007$ and a country-permutation $p = 0.27$; half of its variance lies between countries within regions, more than for any other indicator. It coincides with the general-practitioner tagging

artefact: the median FUA has 10.8 mapped GPs per 100k inhabitants, but the country medians are 2.1 in Sweden, 2.3 in Portugal, 2.6 in Finland and 3.5 in Italy against 24.5 across the West, because primary care there sits in health centres OpenStreetMap tags differently, which overstates everyday deprivation levels and compresses their within-city spread. Mapped GP density is uncorrelated with city size (log-log slope +0.09, country-clustered $p = 0.29$). Eight cities have only one of the two emergency services mapped (Athina, Brăila, Helsinki, Lahti, Norrköping, Szeged, Talavera de la Reina, Žilina); the composite renormalises weights over the services present, their median mean emergency cost is 0.197 against 0.169 for the rest, and none is in the desert group.

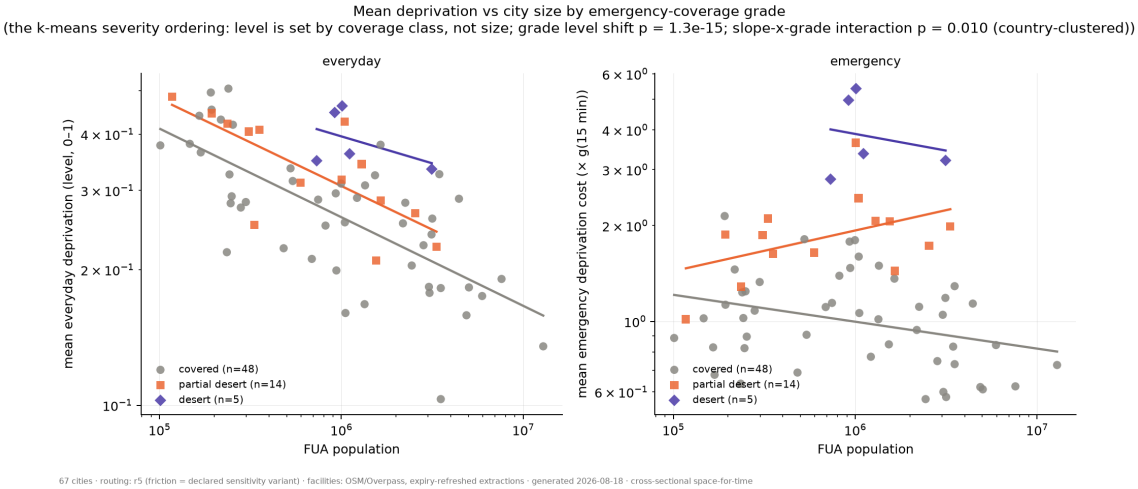


Figure S3: Mean deprivation against city size by emergency-coverage grade, per regime (left: everyday level; right: emergency cost). The grade shifts the emergency level (country-clustered Wald $p \approx 1.3 \times 10^{-15}$; median 1.02 / 1.87 / 3.37); the slope-by-grade interaction is $p = 0.010$ under the current 48/14/5 grades but $p = 0.17$ under the pre-refresh 50/12/5 grades, so it is boundary-sensitive and not headlined. The boundary cities are Brăila and Łomża. Within-grade fits are plain OLS over few cities.

Table S5: Nested decomposition of the variance of each city indicator across the 67 cities: share of the total sum of squares lying between the four macro-regions, between countries within regions, and between cities within countries. Source: `regional_variance_decomposition.csv` (`paper/analysis_regional_structure.py`).

Indicator	Between regions	Between countries (within region)	Within countries	Country level (regions + countries)
coupling ρ	0.20	0.22	0.58	0.42
everyday Gini	0.24	0.51	0.25	0.75
emergency Gini	0.35	0.35	0.30	0.70
divergence gap	0.32	0.44	0.24	0.76
HH share	0.22	0.18	0.60	0.40
compounding intensity	0.21	0.19	0.60	0.40
mean emergency cost	0.25	0.54	0.22	0.78

Table S6: Dispersion of each indicator by macro-region (standard deviation across cities) with a Fligner–Killeen test of equal dispersion at the city level (anti-conservative reference) and the country-permutation p-value of the same statistic (whole countries permuted across regions, 10,000 permutations). Source: `regional_dispersion.csv`.

Indicator	sd North	sd West	sd South	sd CEE	p city	p_{perm}
coupling ρ	0.144	0.102	0.187	0.154	0.143	0.163
everyday Gini	0.075	0.079	0.102	0.060	0.067	0.165
emergency Gini	0.058	0.065	0.076	0.130	0.006	0.028
divergence gap	0.074	0.072	0.102	0.159	0.008	0.010
HH share	0.038	0.026	0.044	0.040	0.278	0.241
compounding intensity	0.017	0.013	0.023	0.020	0.175	0.190
mean emergency cost	0.481	0.297	0.679	1.294	0.035	0.063

Table S7: Regional contrasts re-tested without the five emergency-desert capitals (62 cities, 24 countries): ANOVA on country means and the country-permutation p-value, with the Central-Eastern European and rest-of-sample medians. Source: `inference_regional_no_deserts.csv`.

Indicator	p country ANOVA	p_{perm}	Median CEE	Median rest
coupling ρ	0.0028	0.0030	0.583	0.384
everyday Gini	0.2486	0.2613	0.522	0.458
emergency Gini	0.0080	0.0060	0.507	0.437
divergence gap	0.0364	0.0346	-0.004	-0.054
HH share	0.0012	0.0016	0.359	0.319
compounding intensity	0.0021	0.0023	0.397	0.374
mean emergency cost	0.0170	0.0198	1.441	1.062

Table S8: Coverage grade by macro-region (number of cities). Source: `coverage_grade_by_region.csv`.

Region	Covered	Partial desert	Desert
North	6	7	0
West	15	1	0
South	17	1	0
CEE	10	5	5

Table S9: The 18 Southern cities ordered by everyday Gini: the Spanish and Greek cities lie above the Italian and Portuguese ones with no overlap. Source: `south_everyday_gini_by_country.csv`.

City	CC	Population	G_{ev}	$\bar{D}_{\text{ev}}$
Athina	EL	3,522,264	0.609	0.103
Bilbao	ES	1,053,444	0.570	0.160
Barcelona	ES	5,945,709	0.545	0.175
Jaén	ES	249,477	0.543	0.291
Madrid	ES	7,601,394	0.533	0.191
Alicante	ES	996,867	0.530	0.311
Talavera de la Reina	ES	101,050	0.522	0.378
Palma de Mallorca	ES	818,202	0.503	0.251
Roma	IT	4,432,845	0.424	0.287
Arezzo	IT	146,788	0.412	0.381
Cosenza	IT	239,305	0.407	0.505
Milano	IT	5,030,024	0.400	0.183
Vicenza	IT	280,054	0.391	0.275
Palermo	IT	1,006,055	0.383	0.317
Lisboa	PT	3,132,904	0.367	0.239
Napoli	IT	3,457,649	0.305	0.326
Caserta	IT	298,402	0.298	0.283
Porto	PT	1,531,690	0.272	0.324

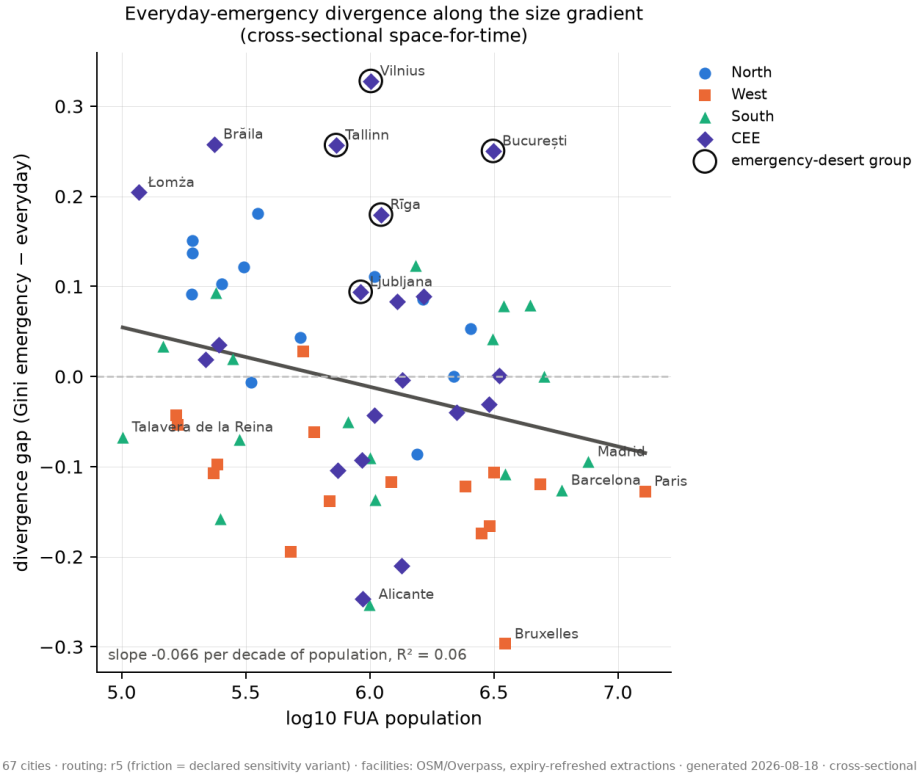


Figure S4: The divergence gap (emergency Gini minus everyday Gini) against $\log_{10}$ FUA population. The gap falls weakly with size (-0.066 per decade, $p = 0.012$, $R^2 = 0.06$); the coupling ρ ($p = 0.14$) and the compounding share ($p = 0.06$) show no significant size gradient.

S5 Compounding: coupling, threshold sensitivity and spatial anatomy

The "how high is high" percentile cut moves the HH share by 29.8 percentage points at the median city, against 1.4 under the whole deprivation-curvature grid and 0.9 under the functional-form swaps. The continuous compounding intensity has fixed anchors (1/3 under independence, 1/2 under perfect coupling, 1/4 under perfect divergence) and differs regionally ($p_{\text{perm}} = 0.0023$, strongest in Central-Eastern Europe). Maps are read under two rules: legends carry population-weighted shares while the picture is area-weighted, and no single cell's class is a finding, since roughly 24 % of population flips class between routing engines.

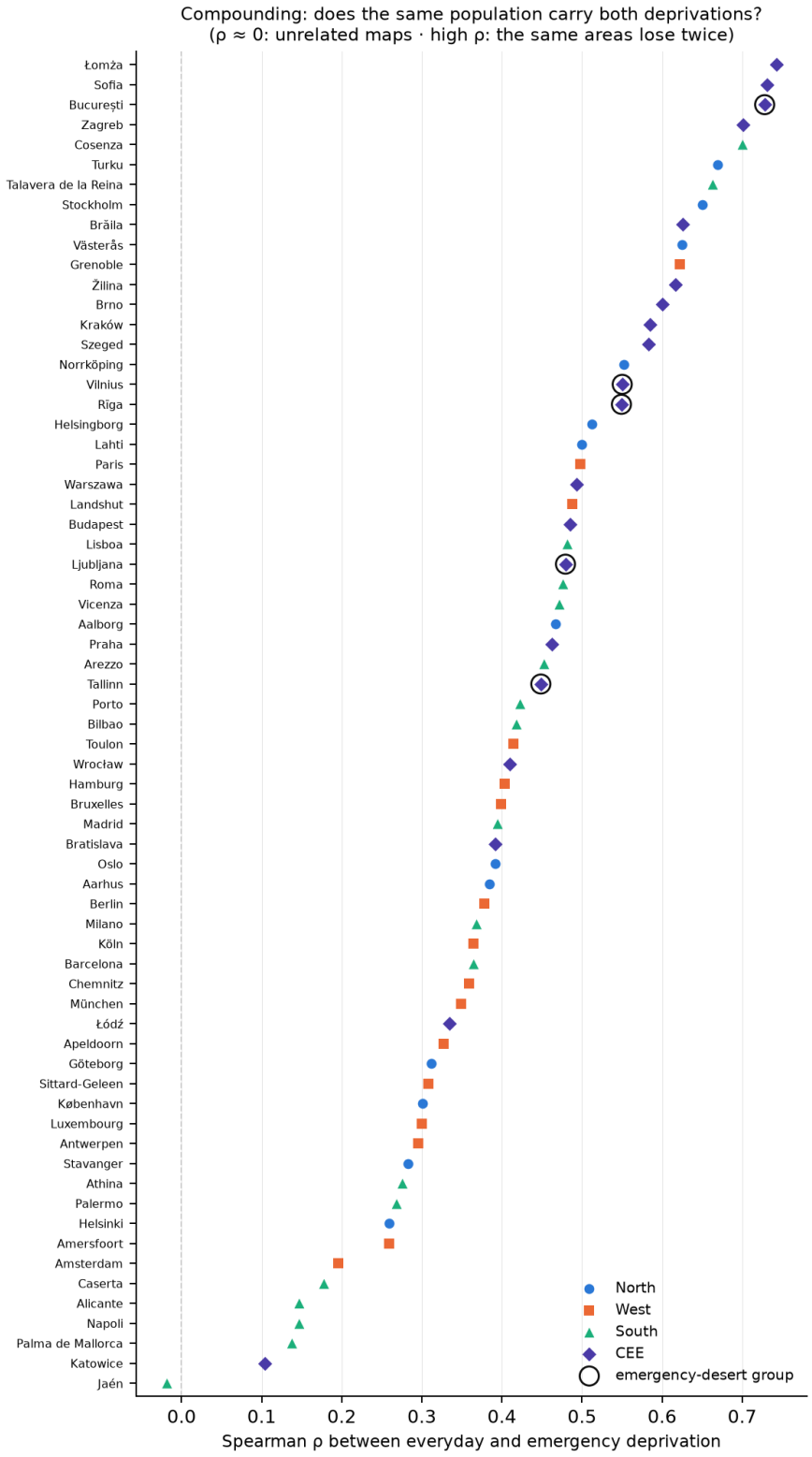


Figure S5: Coupling of the two deprivation surfaces across the 67 cities: Spearman ρ between the everyday and emergency percentile surfaces, cities ranked (-0.02 to 0.74). Colours mark macro-regions; the emergency-desert capitals are outlined. The HH share exceeds the 25 % independence benchmark in 66 of 67 cities (mean 33 %).

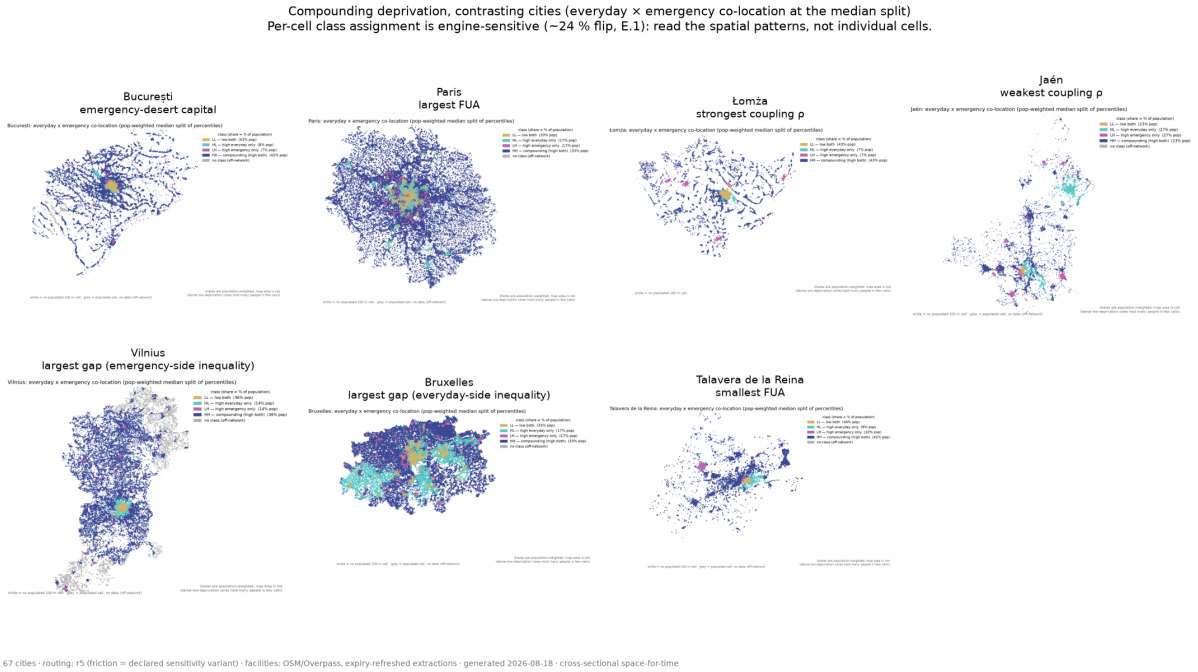


Figure S6: Compounding gallery: median-split co-location typology for contrasting cities (all 67 maps in Section S10). Colours: LL both low, HL high everyday only, LH high emergency only, HH compounding. Legend shares are population-weighted; the maps are area-weighted.

S6 Robustness of the deprivation layer

What is varied is drawn in main-text Fig. 1: the Layer-1 curvature grid ($k \in \{0.1, 0.2, 0.3\}$ per minute $\times$ $t_0 \in \{10, 15, 20\}$ minutes for the everyday logistic; $\lambda \in \{1.4, 1.8, 2.2\}$ for the emergency Box-Cox) and the Layer-2 form swaps calibrated to the same anchors. How far the results move is in Table S10. Levels are not robust, and are not meant to be: curvature alone moves a typical city's everyday Gini by 0.24 Gini points, which is why the calibration carries information and why every slope claim is defended by the specification curve. Ranks and classes are robust: the same grid moves the compounding share by 1.4 percentage points at the median, the form swaps by 0.9, against 29.8 for the threshold. City-level Gini orderings survive with two scoped exceptions: rank agreement with the baseline stays at $\rho \geq 0.90$ over the curvature grid and the everyday form swap, while both emergency escalation swaps reorder the emergency-Gini ranking (exponential $\rho = 0.19$; bounded survival $\rho = 0.49$) because each changes what that Gini measures. Deprivation-variant flip cells average 18.1 % of city population (range 5.2–53.0 %), with the HH class share moving 2.1 percentage points on average and 6.1 at most. The clustering survives its feature-set choice (main partition ARI 1.00, peeled 0.97). The routing-engine cross-check bounds the engine's leverage: switching engines moves travel-time levels by 34–314 % and flips the class of roughly 24 % of population per city, while aggregate class shares move at most 0.7 percentage points.

Table S10: Sensitivity envelopes: median min–max envelope width across the 67 cities per sweep axis and target. The form-swap emergency-Gini envelope spans the three beyond-cut-off escalation hypotheses. Source: `deprivation_sensitivity_summary.csv`.

Sweep axis	Everyday Gini	Emergency Gini	HH share
Curvature (Layer 1)	0.244	0.170	0.014
Form swap (Layer 2)	0.035	0.649	0.009
"How high is high" threshold	–	–	0.298

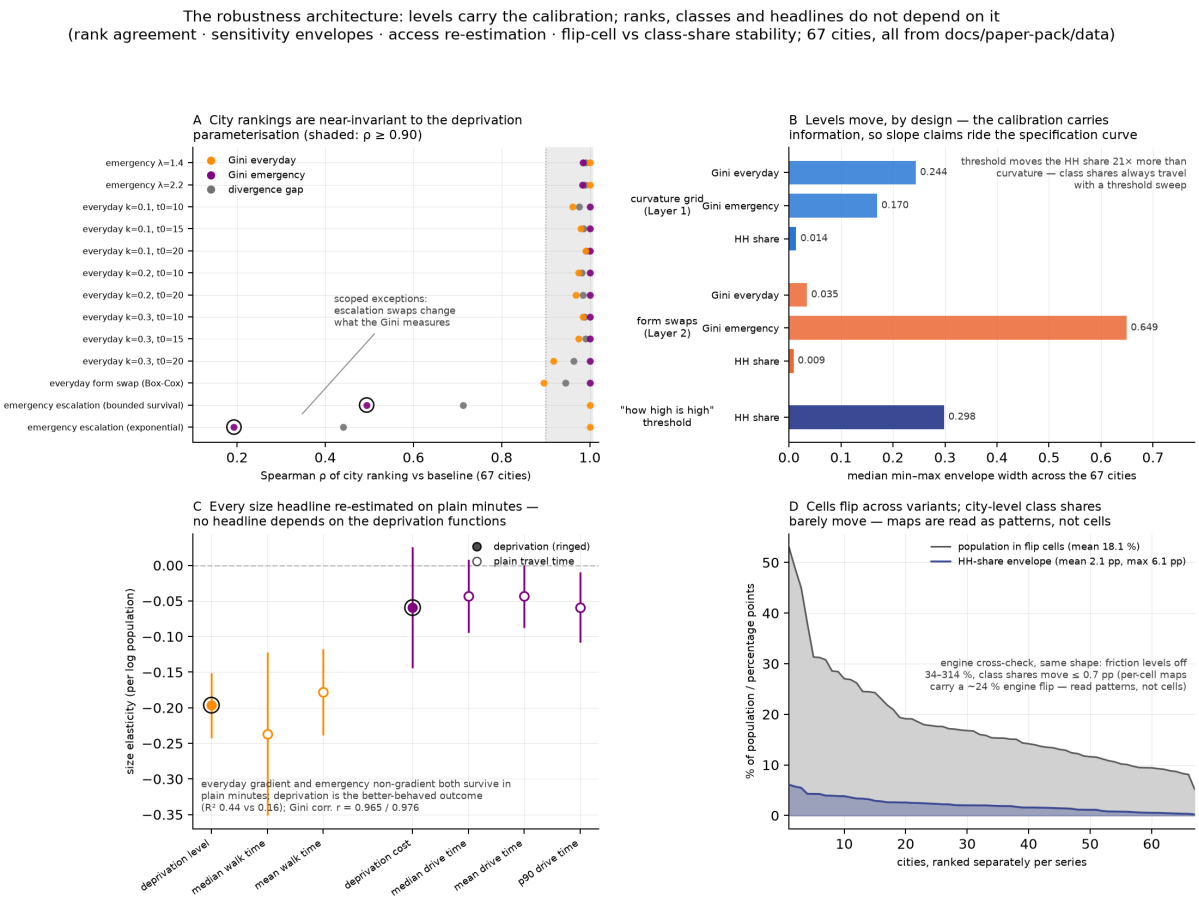


Figure S7: The robustness architecture in one view: rank agreement across deprivation variants, envelope widths per axis, access-based re-estimation of every headline, and flip-cell against class-share stability, with the routing-engine cross-check alongside.

S7 Validation exercises

Routing-engine cross-check. Re-running Hamburg and Köln under the friction-surface engine against the R5 baseline shows travel-time levels understated by 34–314 % with city-specific Gini bias (emergency Gini −30 % in Hamburg, −18.5 % in Köln), which demoted friction to a sensitivity variant. Aggregate typology class shares move ≤ 0.7 pp and the coupling ρ by ≤ 11 %, while roughly 24 % of population flips per-cell class.

OpenStreetMap completeness. Benchmark against national registries where available: Germany, pharmacies ratio 0.976, hospitals 1.24 (the hospital ratio carries a verify-before-publication flag). The per-city facility counts provide the cross-country completeness picture and surface the GP tagging artefact.

External benchmark. The German INKAR accessibility indicators reproduce the study's German pharmacy and supermarket access patterns; no comparable emergency-side benchmark is available.

Re-anchor verification. The emergency reporting anchor moved from $g(45)$ to $g(15)$ in August 2026. For the three cities re-seeded from stored artifacts every rank-based indicator is identical to machine precision and the level ratio equals $g(45)/g(15) = 6.7307$ to 10^{-15}; the other 64 cities were byte-identical between the two runs.

S8 Indicator glossary

Table S11: Compact glossary of the indicators used in the article.

Term	Meaning
FUA	Functional urban area (Eurostat/OECD): city plus commuting zone; the unit "city".
Everyday regime	Five walkable service categories (GP, pharmacy, supermarket, school, green space); deprivation is a bounded *level* in $[0, 1]$.
Emergency regime	ED hospital and ambulance station, reached by car; deprivation is an unbounded convex *cost* in multiples of g(15).
$\bar{D}_{\text{ev}}$, $\bar{C}_{\text{em}}$	Population-weighted mean deprivation per regime; scales are incomparable across regimes.
G_{ev}, G_{em}	Population-weighted within-city Gini of each surface.
Divergence gap	$G_{\text{em}} - G_{\text{ev}}$; positive = emergency-side inequality problem.
ρ	Spearman correlation of the two population-weighted percentile surfaces.
LL/LH/HL/HH	Median-split co-location classes; HH = compounding; independence = 25 %.
Compounding intensity	Population-weighted mean of min(everyday pct, emergency pct); 1/3 independence, 1/2 coupled, 1/4 divergent.
Coverage grade	Severity ordering from the two clustering passes: covered / partial desert / desert.
Elasticity	Slope of log outcome on log population (space-for-time).
Wild cluster bootstrap	Country-clustered resampling; the citable p for scaling claims.
p_{perm}	Permutation p-value reshuffling whole countries across macro-regions.
TOST	Two one-sided equivalence tests; positive evidence a slope lies within a bound.
Specification curve	Headline regression re-estimated under every deprivation parameterisation.
Flip cells	Cells whose typology class changes under any deprivation variant.

S9 Per-city table

Table S12: The 67 functional urban areas, largest first. Grade: emergency-periphery coverage grade (Section S4); n_{em}: number of mapped emergency services (2 = both ED hospital and ambulance station); $\bar{D}_{\mathrm{ev}}$: population-weighted mean everyday deprivation level (0–1); $\bar{C}_{\mathrm{em}}$: mean emergency deprivation cost in multiples of the 15-min benchmark cost $g(15)$; G: population-weighted Gini per regime; ρ: Spearman coupling of the two percentile surfaces; HH: compounding population share at the median split.

City	CC	Region	Population	Grade	n_{em}	$\bar{D}_{\mathrm{ev}}$	$\bar{C}_{\mathrm{em}}$	G_{ev}	G_{em}	ρ	HH
Paris	FR	West	12,886,280	covered	2	0.135	0.73	0.625	0.497	0.50	0.334
Madrid	ES	South	7,601,394	covered	2	0.191	0.62	0.533	0.439	0.39	0.316
Barcelona	ES	South	5,945,709	covered	2	0.175	0.84	0.545	0.418	0.36	0.309
Milano	IT	South	5,030,024	covered	2	0.183	0.61	0.400	0.400	0.37	0.319
Berlin	DE	West	4,875,655	covered	2	0.158	0.62	0.558	0.438	0.38	0.317
Roma	IT	South	4,432,845	covered	2	0.287	1.14	0.424	0.503	0.48	0.342
Athina	EL	South	3,522,264	covered	1	0.103	0.73	0.609	0.500	0.28	0.274
Bruxelles	BE	West	3,521,497	covered	2	0.182	1.29	0.583	0.286	0.40	0.332
Napoli	IT	South	3,457,649	covered	2	0.326	0.83	0.305	0.384	0.15	0.270
Warszawa	PL	CEE	3,334,987	partial	2	0.225	1.99	0.589	0.590	0.49	0.347
Hamburg	DE	West	3,162,607	covered	2	0.260	0.58	0.543	0.437	0.40	0.321
Bucureşti	RO	CEE	3,140,502	desert	2	0.335	3.21	0.524	0.774	0.73	0.424
Lisboa	PT	South	3,132,904	covered	2	0.239	1.19	0.367	0.409	0.48	0.319
München	DE	West	3,054,507	covered	2	0.178	0.60	0.616	0.449	0.35	0.304
Budapest	HU	CEE	3,030,949	covered	2	0.183	1.05	0.538	0.507	0.48	0.327
Amsterdam	NL	West	2,829,893	covered	2	0.227	0.75	0.501	0.326	0.20	0.285
Stockholm	SE	North	2,549,795	partial	2	0.267	1.73	0.460	0.513	0.65	0.389
Köln	DE	West	2,438,338	covered	2	0.204	0.57	0.520	0.398	0.36	0.312
Praha	CZ	CEE	2,247,160	covered	2	0.282	1.11	0.515	0.475	0.46	0.354
København	DK	North	2,180,998	covered	2	0.254	0.94	0.401	0.401	0.30	0.307
Sofia	BG	CEE	1,647,936	partial	2	0.285	1.44	0.511	0.600	0.73	0.408
Helsinki	FI	North	1,642,879	covered	1	0.379	1.36	0.341	0.426	0.26	0.270
Oslo	NO	North	1,550,450	partial	2	0.209	2.06	0.553	0.467	0.39	0.297
Porto	PT	South	1,531,690	covered	2	0.324	0.85	0.272	0.395	0.42	0.321
Kraków	PL	CEE	1,350,286	covered	2	0.308	1.50	0.523	0.519	0.58	0.396
Katowice	PL	CEE	1,341,890	covered	2	0.168	1.02	0.489	0.279	0.10	0.267
Zagreb	HR	CEE	1,293,698	partial	2	0.343	2.07	0.499	0.583	0.70	0.395
Antwerpen	BE	West	1,219,366	covered	2	0.289	0.77	0.523	0.406	0.29	0.311
Rīga	LV	CEE	1,110,303	desert	2	0.362	3.37	0.502	0.682	0.55	0.374
Bilbao	ES	South	1,053,444	covered	2	0.160	1.06	0.570	0.433	0.42	0.324
Łódź	PL	CEE	1,045,886	covered	2	0.255	1.60	0.534	0.491	0.33	0.310
Göteborg	SE	North	1,042,720	partial	2	0.427	2.44	0.335	0.446	0.31	0.298
Vilnius	LT	CEE	1,010,376	desert	2	0.462	5.39	0.411	0.739	0.55	0.358
Palermo	IT	South	1,006,055	partial	2	0.317	3.64	0.383	0.292	0.27	0.289
Alicante	ES	South	996,867	covered	2	0.311	1.80	0.530	0.277	0.15	0.269
Wrocław	PL	CEE	939,076	covered	2	0.199	1.47	0.609	0.362	0.41	0.341
Bratislava	SK	CEE	931,613	covered	2	0.296	1.78	0.522	0.430	0.39	0.336
Ljubljana	SI	CEE	919,303	desert	2	0.447	4.96	0.387	0.481	0.48	0.341
Palma de Mallorca	ES	South	818,202	covered	2	0.251	1.39	0.503	0.453	0.14	0.294
Brno	CZ	CEE	745,397	covered	2	0.286	1.14	0.561	0.457	0.60	0.369
Tallinn	EE	CEE	731,883	desert	2	0.349	2.80	0.456	0.713	0.45	0.326
Grenoble	FR	West	688,170	covered	2	0.211	1.11	0.583	0.444	0.62	0.379
Luxembourg	LU	West	595,677	partial	2	0.312	1.64	0.469	0.407	0.30	0.301
Toulon	FR	West	540,279	covered	2	0.315	0.91	0.431	0.458	0.41	0.326
Aarhus	DK	North	525,545	covered	2	0.336	1.81	0.394	0.438	0.38	0.310
Chemnitz	DE	West	481,607	covered	2	0.223	0.69	0.558	0.363	0.36	0.319
Turku	FI	North	353,992	partial	2	0.409	1.63	0.385	0.565	0.67	0.360
Stavanger	NO	North	331,780	partial	2	0.252	2.10	0.498	0.492	0.28	0.299
Aalborg	DK	North	309,890	partial	2	0.405	1.87	0.419	0.541	0.47	0.342
Caserta	IT	South	298,402	covered	2	0.283	1.33	0.298	0.228	0.18	0.272
Vicenza	IT	South	280,054	covered	2	0.275	1.08	0.391	0.411	0.47	0.344
Helsingborg	SE	North	253,346	covered	2	0.420	0.90	0.284	0.387	0.51	0.366

Table S12 continued

City	CC	Region	Population	Grade	n_{em}	$\bar{D}_{\mathrm{ev}}$	$\bar{C}_{\mathrm{em}}$	G_{ev}	G_{em}	ρ	HH
Jaén	ES	South	249,477	covered	2	0.291	1.24	0.543	0.385	-0.02	0.233
Szeged	HU	CEE	246,224	covered	1	0.281	0.82	0.533	0.568	0.58	0.392
Apeldoorn	NL	West	242,638	covered	2	0.326	1.02	0.439	0.341	0.33	0.333
Cosenza	IT	South	239,305	covered	2	0.505	1.23	0.407	0.500	0.70	0.390
Brăila	RO	CEE	235,919	partial	1	0.422	1.29	0.434	0.692	0.63	0.359
Amersfoort	NL	West	234,650	covered	2	0.219	0.64	0.458	0.350	0.26	0.282
Žilina	SK	CEE	217,546	covered	1	0.431	1.46	0.424	0.443	0.62	0.365
Västerås	SE	North	193,510	covered	2	0.454	1.13	0.327	0.478	0.62	0.387
Norrköping	SE	North	193,317	partial	1	0.445	1.88	0.423	0.560	0.55	0.353
Lahti	FI	North	191,532	covered	1	0.495	2.14	0.353	0.444	0.50	0.345
Landshut	DE	West	168,617	covered	2	0.364	0.68	0.492	0.438	0.49	0.361
Sittard-Geleen	NL	West	165,544	covered	2	0.440	0.83	0.316	0.273	0.31	0.287
Arezzo	IT	South	146,788	covered	2	0.381	1.02	0.412	0.446	0.45	0.342
Łomża	PL	CEE	117,255	partial	2	0.484	1.01	0.417	0.621	0.74	0.430
Talavera de la Reina	ES	South	101,050	covered	1	0.378	0.89	0.522	0.454	0.66	0.415

S10 Per-city compounding maps

The median-split co-location typology map of each of the 67 cities, ordered by population. Legends carry population-weighted class shares; the maps are area-weighted. Read for spatial pattern only.

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[1] Adam Wagstaff, Pierella Paci, and Eddy van Doorslaer. On the measurement of inequalities in health. *Social Science & Medicine*, 33(5):545–557, 1991. doi: 10.1016/0277-9536(91)90212-U.

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[3] Lawrence Hubert and Phipps Arabie. Comparing partitions. *Journal of Classification*, 2: 193–218, 1985. doi: 10.1007/BF01908075.

[4] José Holguín-Veras, Noel Pérez, Miguel Jaller, Luk N. Van Wassenhove, and Felipe Aros-Vera. On the appropriate objective function for post-disaster humanitarian logistics models. *Journal of Operations Management*, 31(5):262–280, 2013. doi: 10.1016/j.jom.2013.06.002.

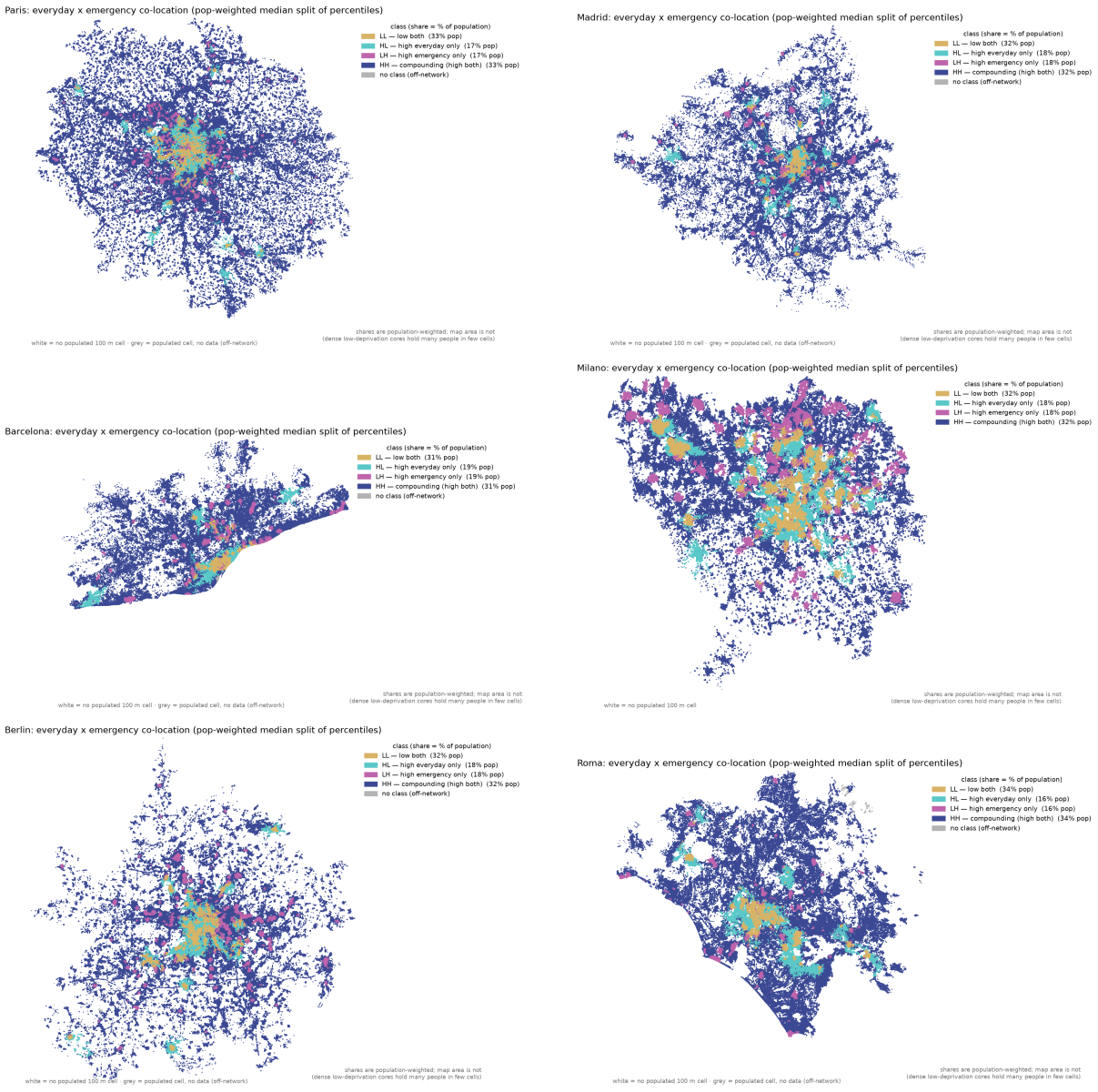


Figure S8: Per-city compounding maps (median-split co-location typology), cities 1–6 by population: Paris (FR), Madrid (ES), Barcelona (ES), Milano (IT), Berlin (DE), Roma (IT). Read for spatial pattern only; per-cell classes carry a $\sim$24 % routing-engine flip share (E.1).

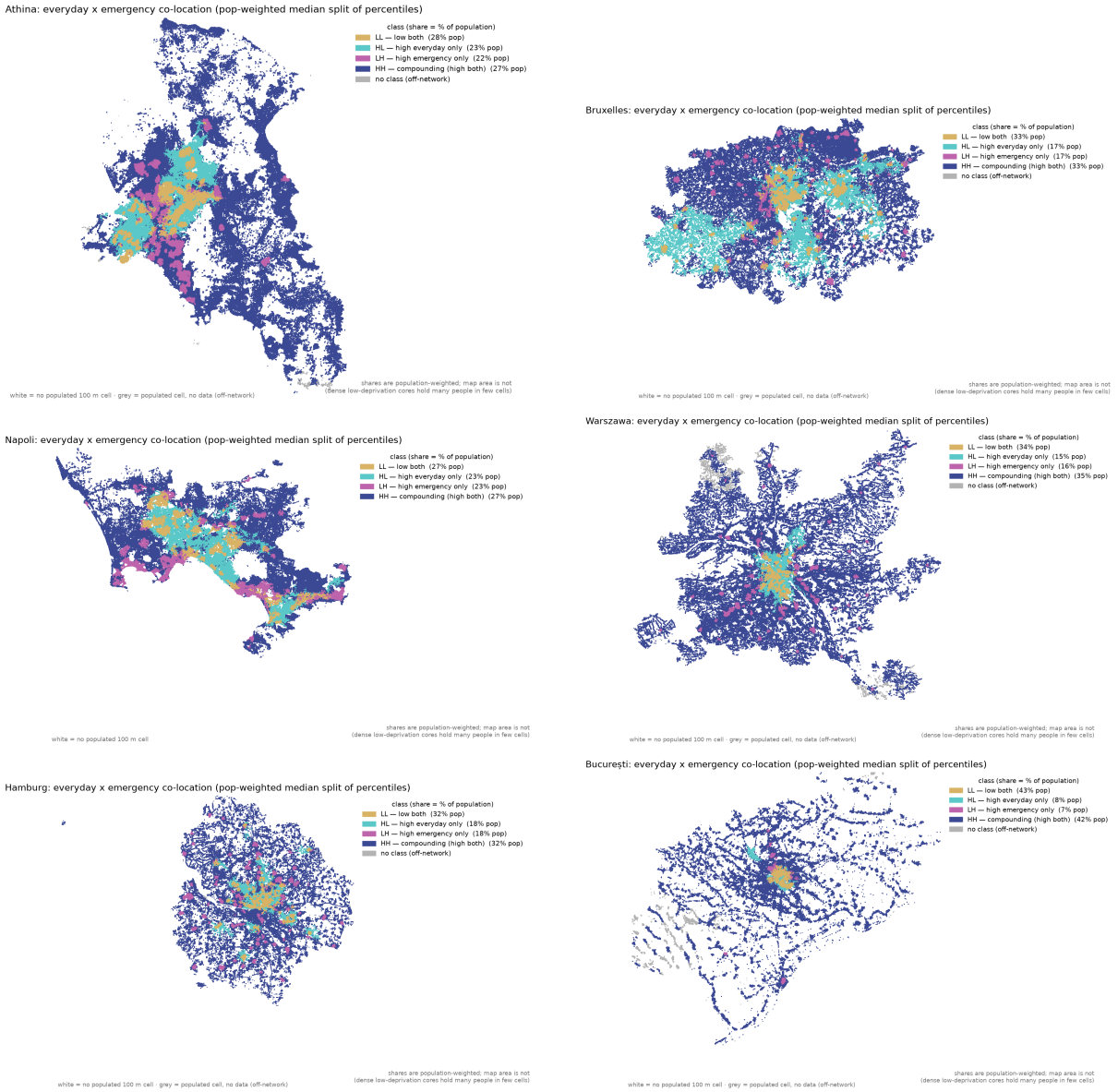


Figure S9: Per-city compounding maps (median-split co-location typology), cities 7–12 by population: Athina (EL), Bruxelles (BE), Napoli (IT), Warszawa (PL), Hamburg (DE), Bucureşti (RO). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

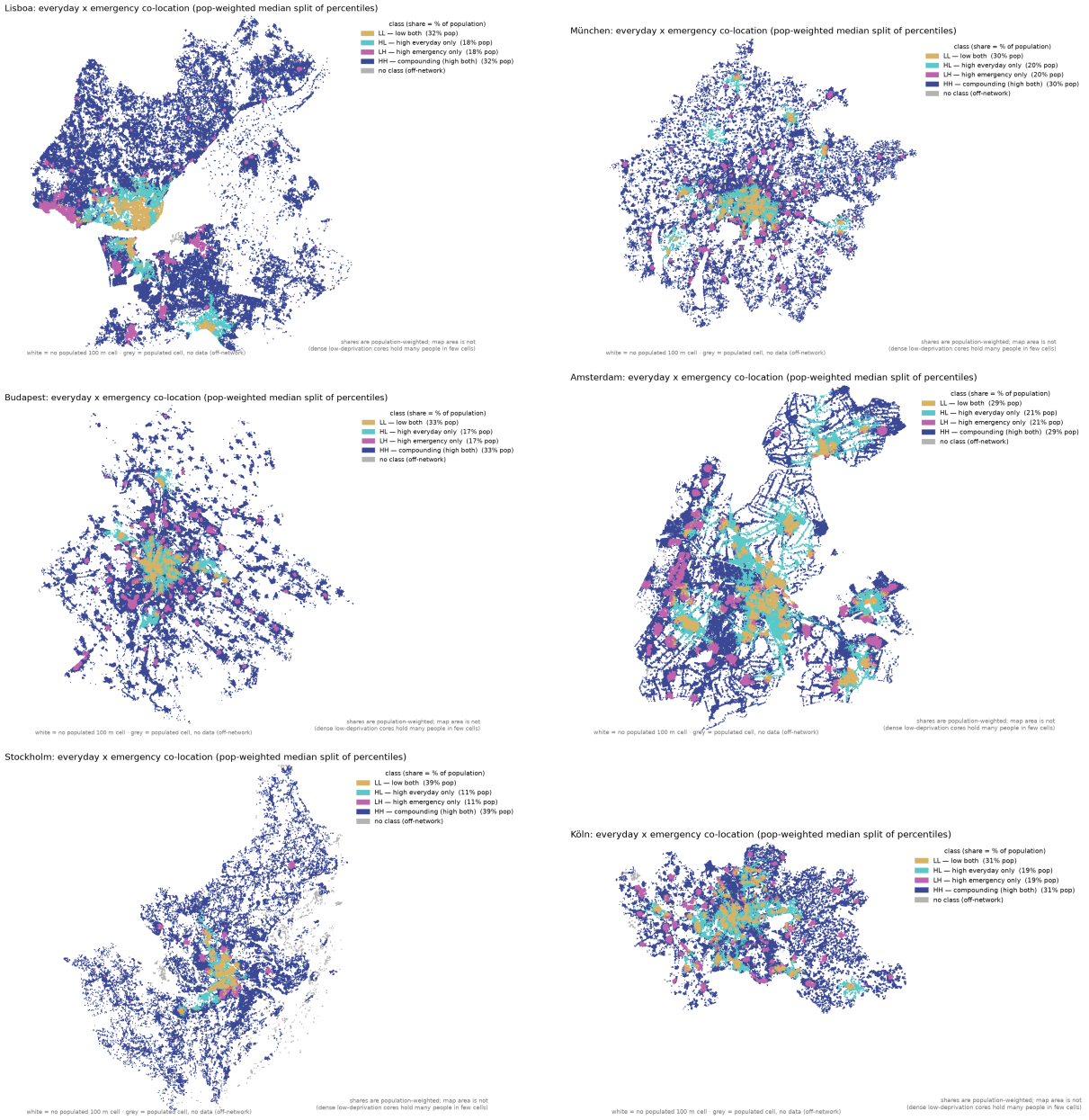


Figure S10: Per-city compounding maps (median-split co-location typology), cities 13–18 by population: Lisboa (PT), München (DE), Budapest (HU), Amsterdam (NL), Stockholm (SE), Köln (DE). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

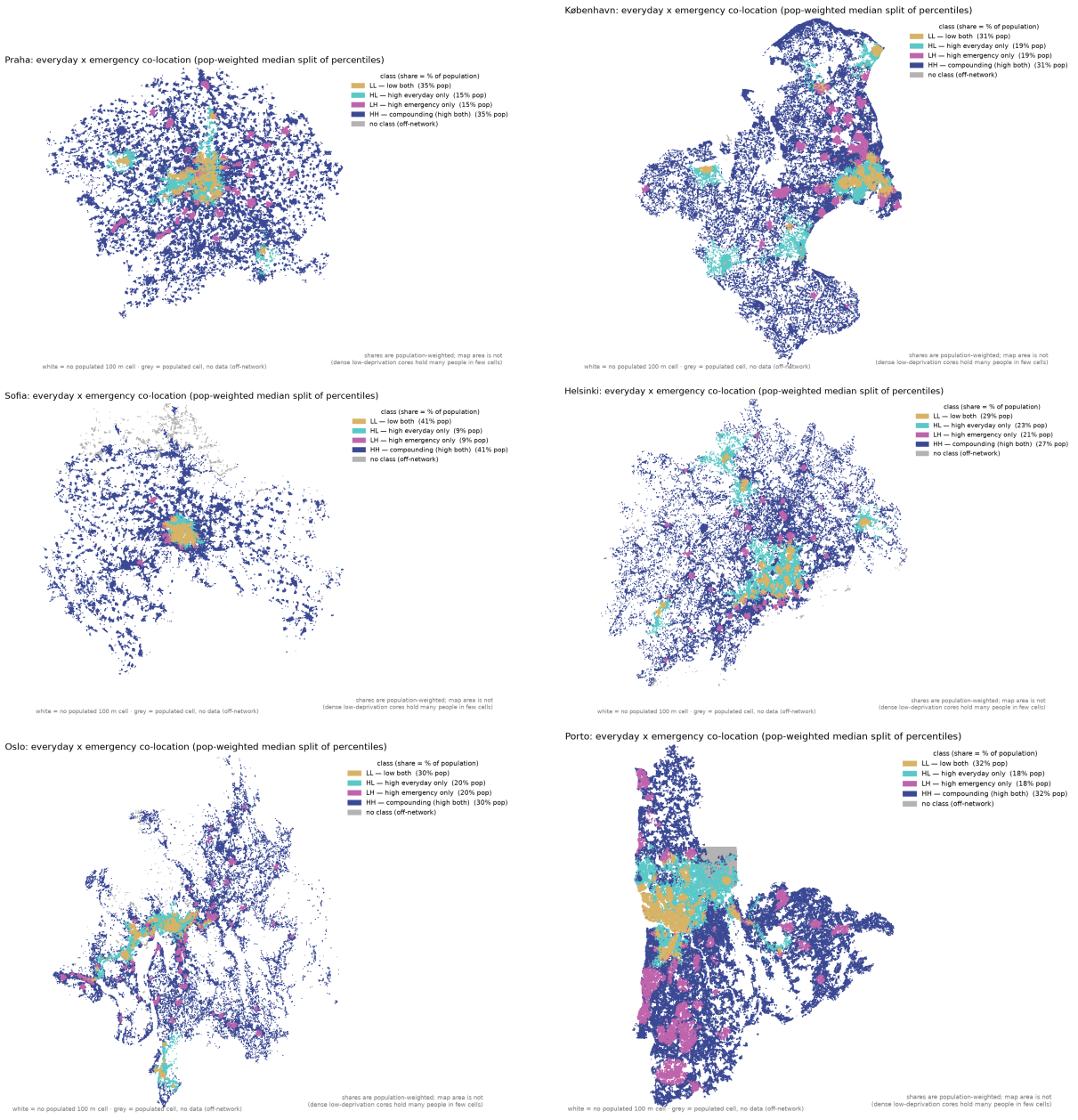


Figure S11: Per-city compounding maps (median-split co-location typology), cities 19–24 by population: Praha (CZ), København (DK), Sofia (BG), Helsinki (FI), Oslo (NO), Porto (PT). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

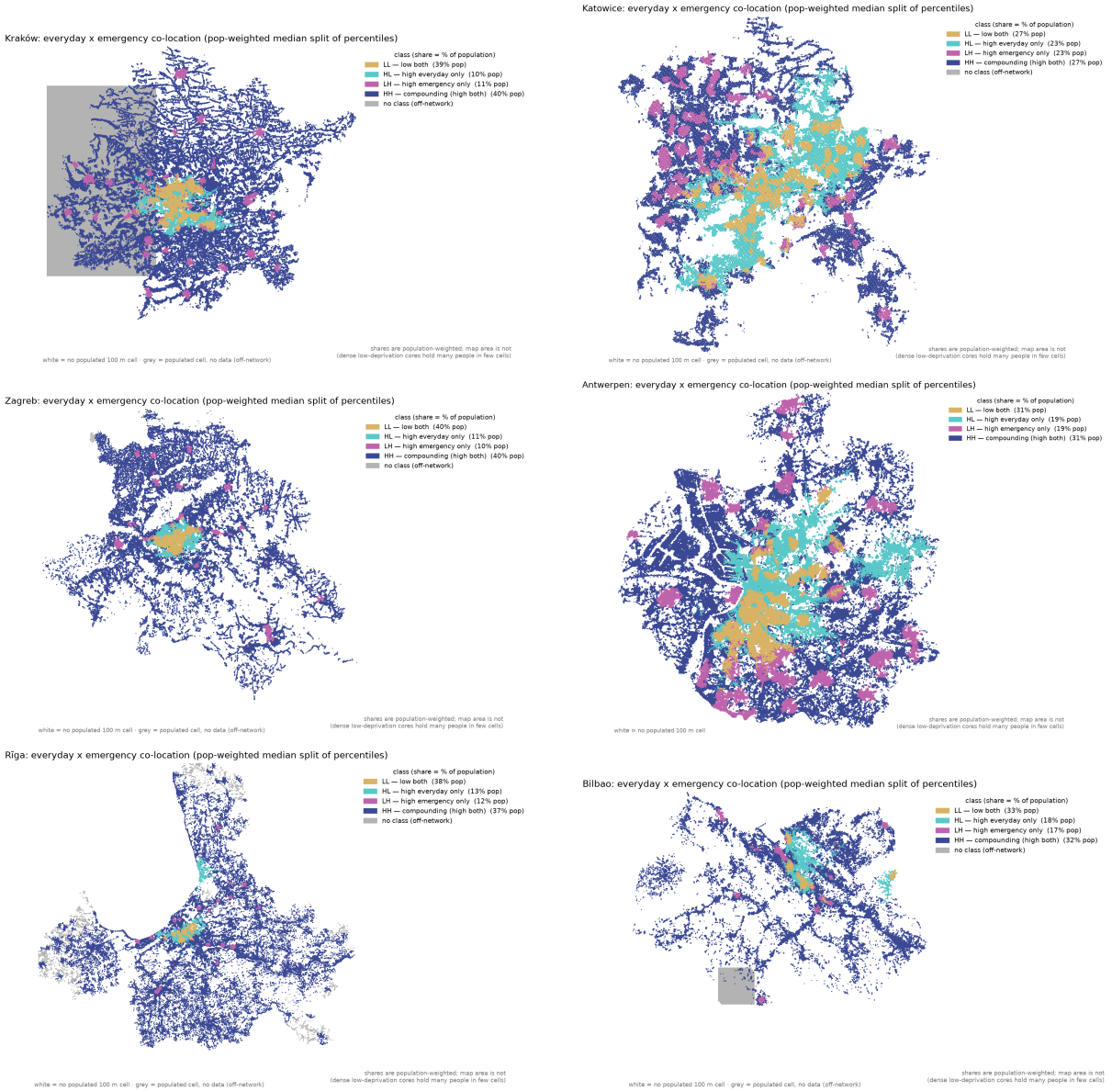


Figure S12: Per-city compounding maps (median-split co-location typology), cities 25–30 by population: Kraków (PL), Katowice (PL), Zagreb (HR), Antwerpen (BE), Rīga (LV), Bilbao (ES). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

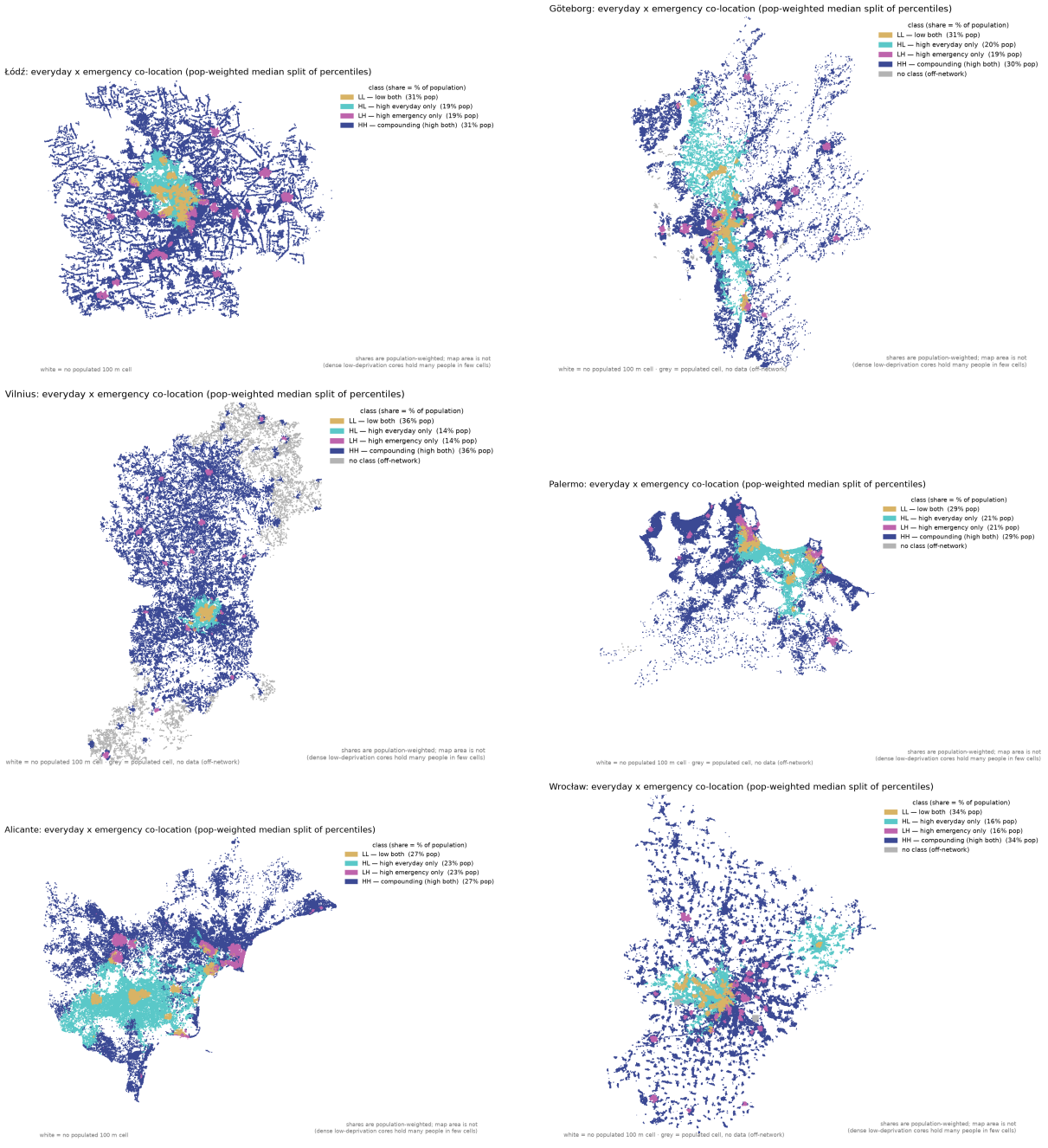


Figure S13: Per-city compounding maps (median-split co-location typology), cities 31–36 by population: Łódź (PL), Göteborg (SE), Vilnius (LT), Palermo (IT), Alicante (ES), Wrocław (PL). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).



Figure S14: Per-city compounding maps (median-split co-location typology), cities 37–42 by population: Bratislava (SK), Ljubljana (SI), Palma de Mallorca (ES), Brno (CZ), Tallinn (EE), Grenoble (FR). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

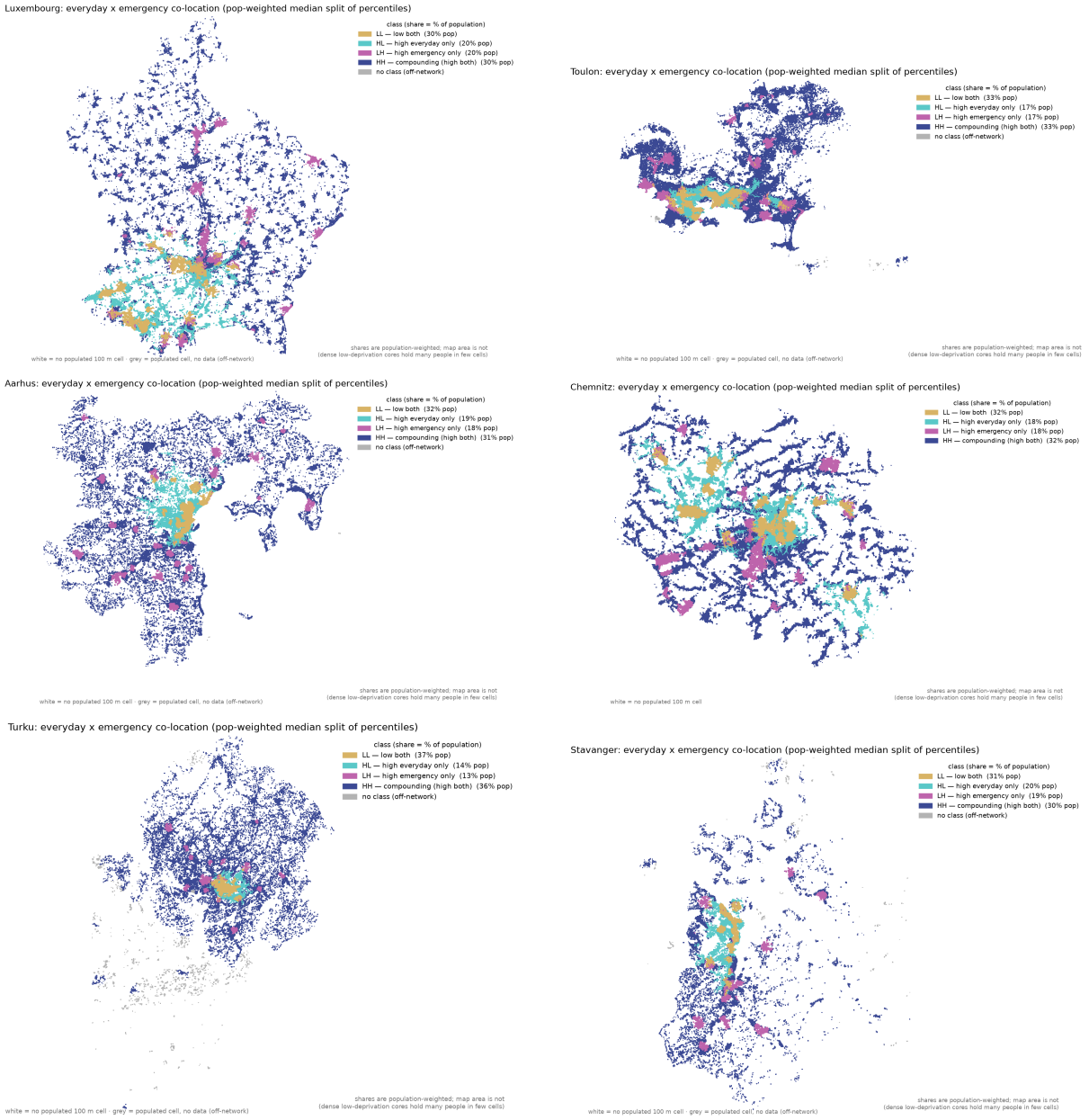


Figure S15: Per-city compounding maps (median-split co-location typology), cities 43–48 by population: Luxembourg (LU), Toulon (FR), Aarhus (DK), Chemnitz (DE), Turku (FI), Stavanger (NO). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

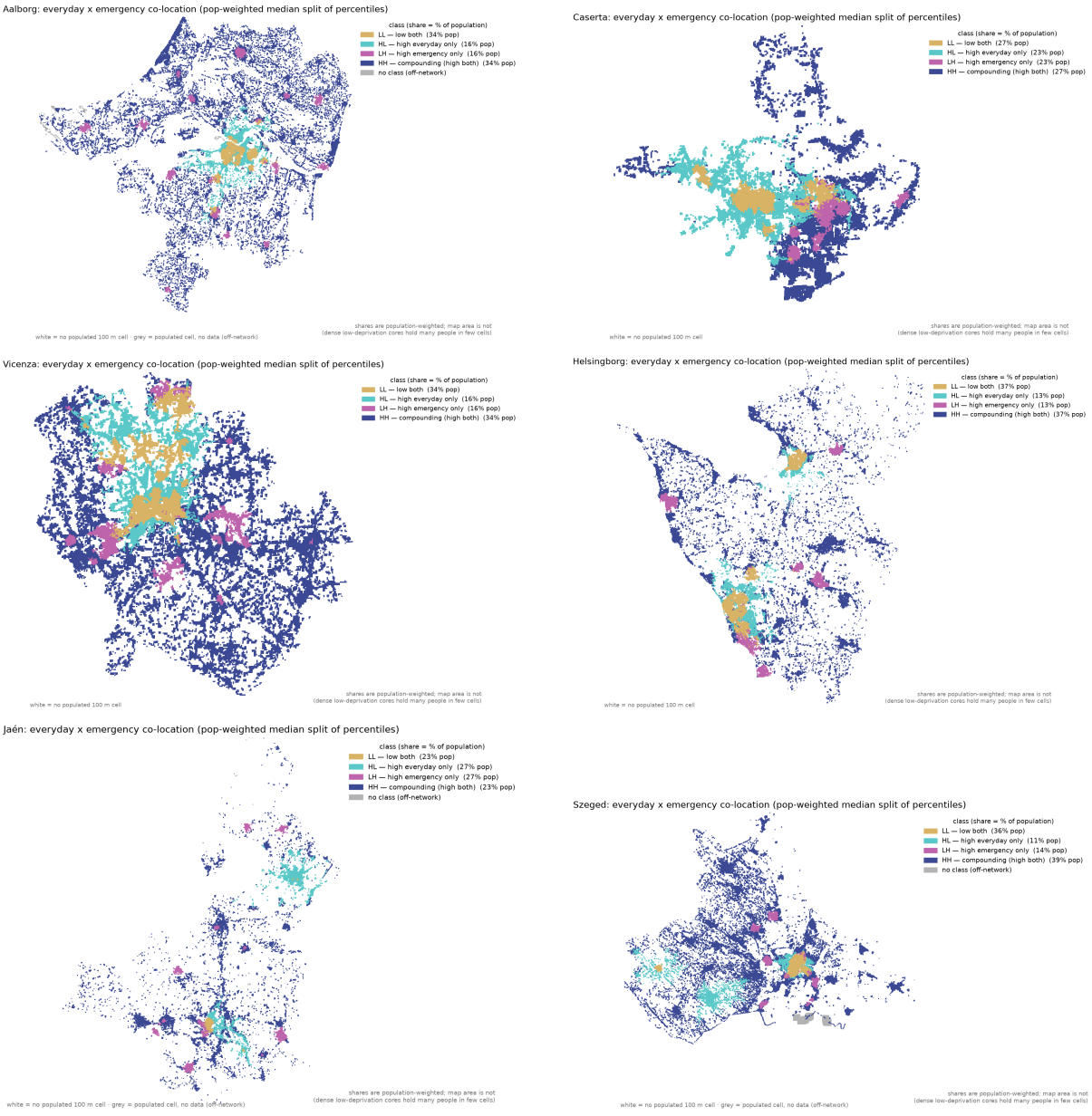


Figure S16: Per-city compounding maps (median-split co-location typology), cities 49–54 by population: Aalborg (DK), Caserta (IT), Vicenza (IT), Helsingborg (SE), Jaén (ES), Szeged (HU). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

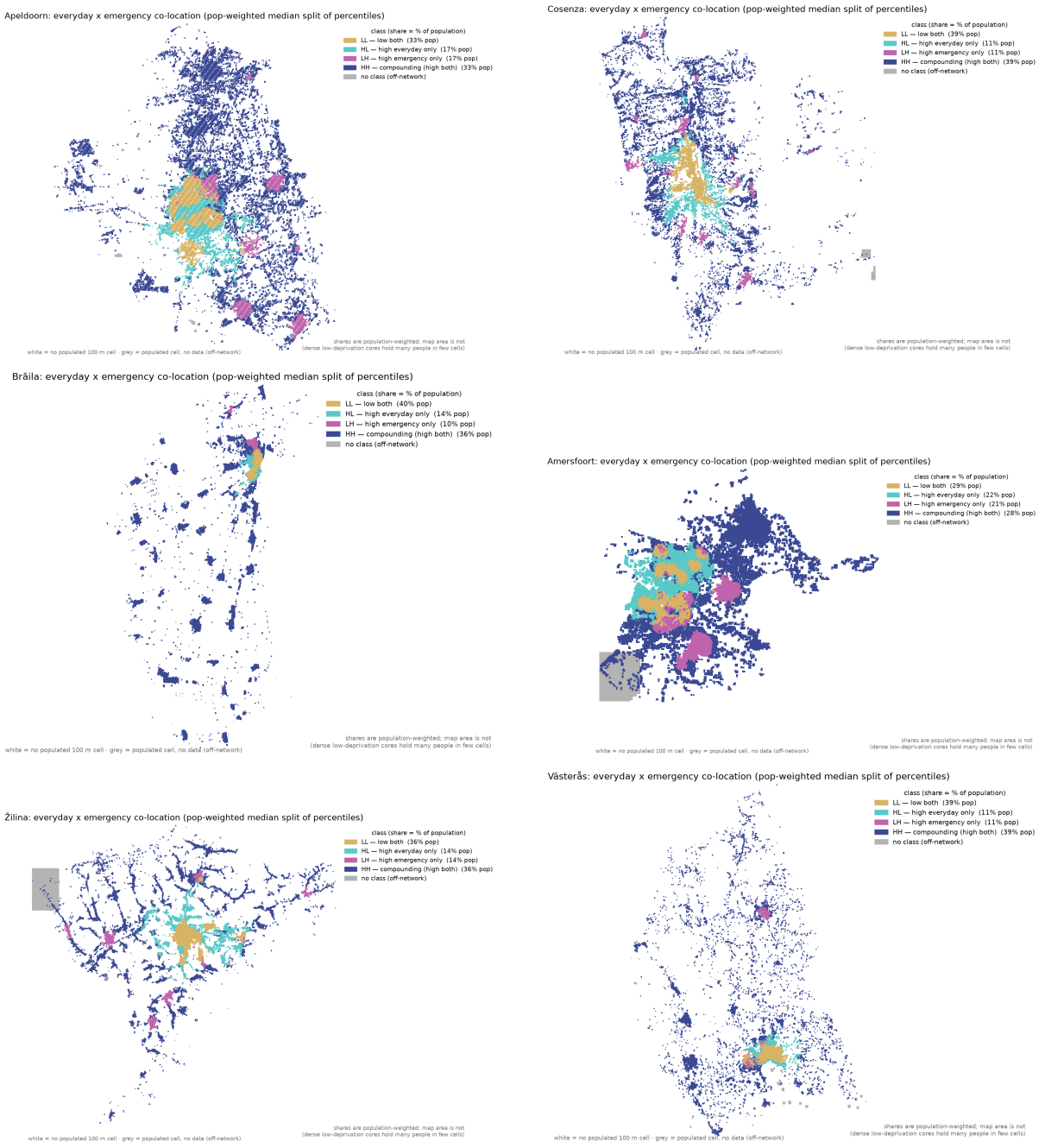


Figure S17: Per-city compounding maps (median-split co-location typology), cities 55–60 by population: Apeldoorn (NL), Cosenza (IT), Brăila (RO), Amersfoort (NL), Žilina (SK), Västerås (SE). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

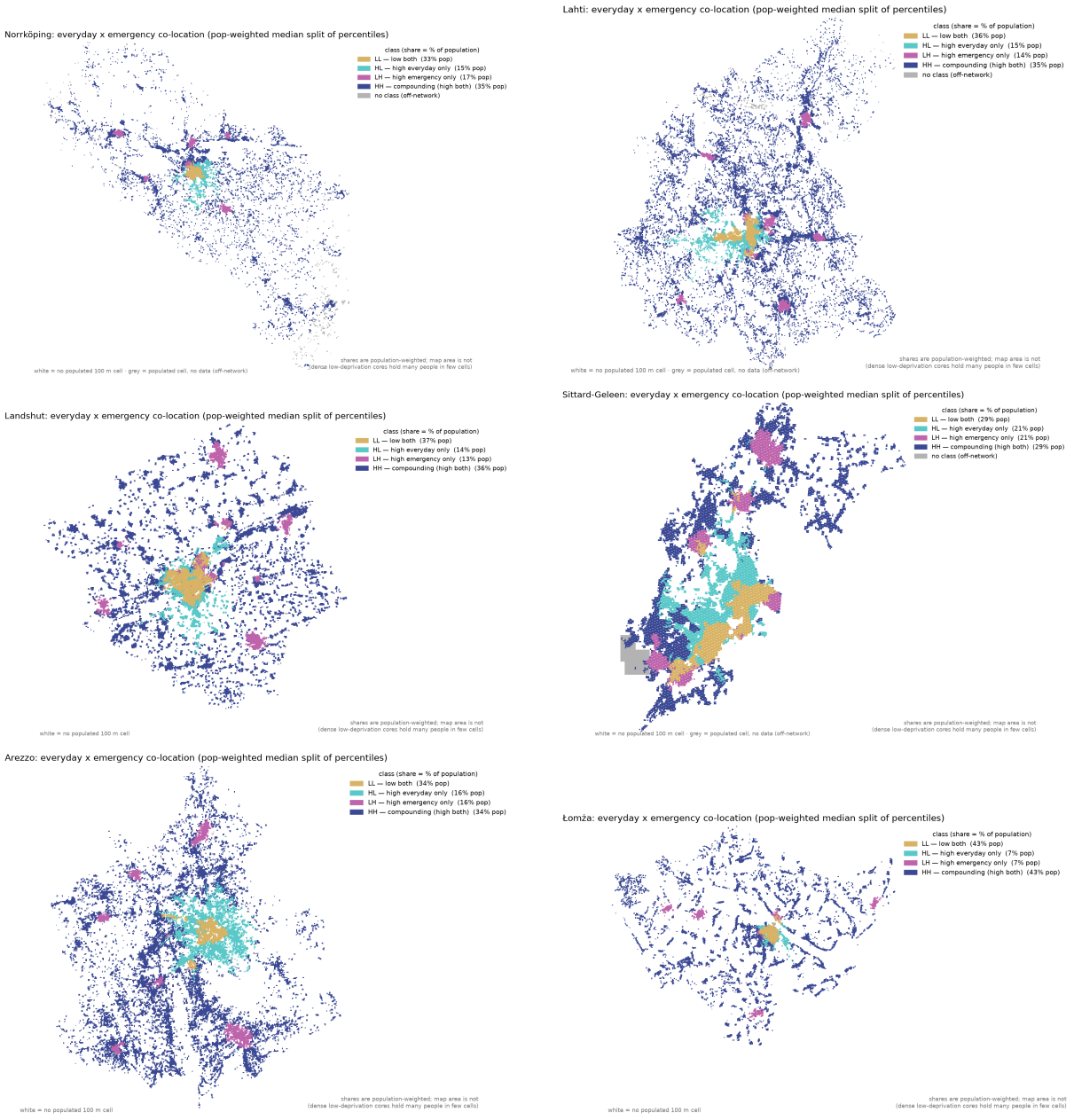


Figure S18: Per-city compounding maps (median-split co-location typology), cities 61–66 by population: Norrköping (SE), Lahti (FI), Landshut (DE), Sittard-Geleen (NL), Arezzo (IT), Łomża (PL). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

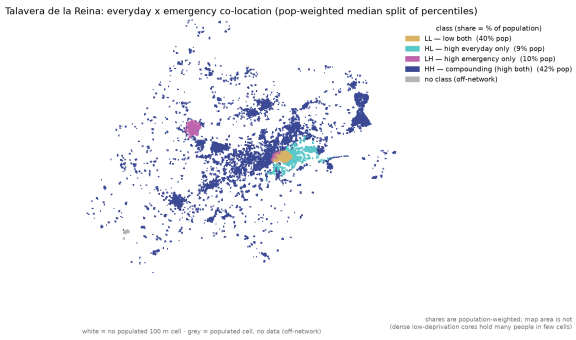


Figure S19: Per-city compounding maps (median-split co-location typology), cities 67–67 by population: Talavera de la Reina (ES). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).